Merge and Principle G

Abstract

To engage in activities such as language, music, arithmetic, metrical stress, chess etc., humans generate complex, novel and unbounded mental representations. Call the mind that generates such representations the *generative mind*. It is plausible to suppose that the generative mind consists of some powerful combinatorial devices that operate under the principle of combining two mental objects in an unbounded fashion: call it *Principle G*.

The features of novelty and unboundedness in the generative devices suggest that the generative mind is human-specific. Hence it is unlikely that they were adopted from pre-hominid species; so we face an explanatory gap for the emergence of each device. We attempt to minimize the explanatory gap by exploring the methodological prospect that a single generative device satisfies Principle G for all domains.

To that end, we study the generative device of language, *Merge*. I argue that the simplest form of Merge may satisfy Principle G throughout the generative mind because (a) there is nothing in the formulation of Merge that makes it language-specific, (b) versions of Merge seem to be operating in non-linguistic domains such as arithmetic and music.

Merge and Principle G

Humans have the impressive ability to generate unbounded structured thoughts and actions from limited resources in a variety of domains. Most children can draw landscapes and animal figures with a few coloured pencils. Millions of children can manipulate a handful of notes to play complex music on various instruments. Every winter, people across the world knit a variety of woollen garments with two knitting needles and some wool. With a finite stock of words, human beings can generate virtually infinite sentences of a language. Once we get a sense of the phenomenon, we can witness this ability in almost everything humans do: cooking, building houses, tailoring, inventing games including very complex yoga postures, shopping, gardening, planning the day, scheduling tours and the like.

Each of the listed abilities consist of a finite system of 'atomic' elements which are combined over and over again in different dimensions and hierarchies to generate extremely complex objects. These objects are essentially unbounded in that, although every object so generated is necessarily finite in space and time, there is a clear intuition that a given complex object may be made even more complex in a variety of direction. One can always expand a given sentence, number, painting, building, cooked item and garment. In that sense, each generated object is unbounded and potentially novel.

Although I have listed a variety of actions such as speaking, painting, cooking, knitting, tailoring and the like, it is clear that these actions are accompanied by structured mental representations that guide the actions in suitable circumstanes. These structured representations result when a handful of initial atomic representations in a domain are repeatedly combined by some combinatorial operation. Correspoding to each domain of action thus there is a mental system that generates these representations for the domain. For convenience, let us call the repository of these representations the *Generative Mind*.[1]

Some of the systems of the generative mind, such as language, music, arithmetic and the like, are under study in various familiar disciplines. In the proposed picture, the generative mind also consists of mental systems for cooking, knitting, playing games and the like which, with a few exceptions, are seldom studied in the cognitive sciences. While keeping an eye on the generative mind in its entirety, this paper will be largely restricted to the well-studied mental systems to enforce some grounding in the existing literature. As noted, each mental system consists of some atomic elements and some structuring operations among other things. In this paper, I will be primarily concerned with these structuring operations to focus on the aspect of generativity of mental systems. The basic inquiry is to form some idea of how these structuring operations are distributed across the mental systems of the generative mind.

To anticipate, I will suggest that, at some level of abstraction, it is plausible to think of the generative devices of mental systems such as language, music, arithmetic and much much else as identical even if the mental systems themselves vary widely in content. I must emphasize that, except for some suggestive examples, the inquiry is primarily conceptual and methodological. Whether a single operation in fact obtains across all mental systems in the generative mind is largely an empirical issue that falls beyond the scope of this paper (see XXX 2022).

Principle G

Prima facie, it seems that there is an underlying sense of unity in the productive part of these systems. The mentioned mental representations no doubt vary widely in terms of their content: linguistic representations are made of words, musical representations consist of tones, pictorial representations consist of visual images, and so on. However, when we abstract away from the content of these representations, the combinatorial operations themselves do not seem to be all that different in that they are all essentially engaged in putting two things together over and over again to generate more complex representations in an unbounded fashion.

To lend some substance to the suggested sense of unity, we can say that each mental system raises what Noam Chomsky (2017) calls the *Galilean challenge* in the context of language-theory. The Galilean challenge is to explain the ability of humans to construct an infinite variety of structured thoughts from a finite stock of words, which themselves are constructed out of a handful of sounds or characters, among other things. Exactly the same explanatory challenge arises for the infinite variety of musical forms that are constructed from a small set of notes across almost every known human culture. Moreover, the infinite series of numbers poses the same challenge since the number series is built up successively from a very small set of numbers, possibly just one as we will see (Chomsky 2007). Let us assume that a similar Galilean challenge obtains for cooking, painting, and other activities where humans are able to create an endless variety of complex items from a finite stock of atomic units.

Thus, the generative mind on the whole appears to consist of a basic principle of generativity under which a mental system meets the Galilean challenge for that system: call it *Principle G*, "G" for generativity. I am calling it a *principle* rather than an *operation* because, at this stage of proposal, Principle G is just an abstract idea that assigns generativity to each mental system. The simplest way of endowing generativity is to take objects X, Y from the store of the system, and form a new object Z. According to Chomsky (2015, 16), this mechanism is embedded in some manner in *every* relevant computational procedure.[2] Principle G just states this feature of any generative system. The task then is to find out which mechanisms or operations satisfy Principle G in a given domain. Ideally, for the simplest explanation, we may suppose that the generative mind consists of a single operation that satisfies Principle G in each domain. Later we will see that the supposition of singularity does seem to have some methodological and empirical support.

However, at this stage of understanding, there does not seem to be much warrant for asserting that Principle G is satisfied by a single operation for all systems. For example, it is sometimes said that the generative mind is closely connected to the human linguistic capacity.[3] The observation could mean that the capacity has broad coverage. No doubt the capacity is centrally involved in various language-related generative abilities. For example, George Striedter (2004, 297) suggests that 'no other animal writes books, lectures on history, or tells stories around a fireplace', each of which involves explicit use of language. Beyond such cases, aspects of human evolution seem to suggest that the capacity might have led to various non-linguistic generative abilities as well. It is reported that during the Upper Palaeolithic explosion, when 'language was everywhere', tools, burials, living sites, and occasional hints of art 'reveal beings capable of planning and foresight and a grasp of symbols' (Holden 1998). Noam Chomsky (2020) suggests that the symbolic objects found in the Blombos Caves in South Africa can be attributed to the widespread presence of language.[4]

However, it is unclear which aspect of language is related to some non-linguistic abilities. Intuitively, there seems to be no direct connection between the ability to speak Warlpiri or French, and other generative activities such as whistling North Indian music and cooking Peking Duck. Recent neurological evidence suggests that various mental functions such as music and arithmetic, which are prime examples of the generative mind, remain intact despite severe impairment in various components of language (Fedorenko and Varley 2016, Fedorenko et al. 2024). So, it is *prima facie* implausible that the emergence of language somehow led to music and arithmetic, not to mention cooking and soccer. Yet Principle G is undoubtedly implicated in each of language, music and arithmetic.

It follows that Principle G at best covers some shared component of otherwise disjoint mental systems. In other words, instead of saying that the language system as a whole gave rise to musical and arithmetical systems as wholes, it is perhaps more plausible to suggest that these are otherwise independent systems with a shared component. Since mental systems consist of atomic elements and generative devices, the shared component can only be some generative device, if at all. This is because mental systems differ widely from each other in their specific atomic units: words for language, notes for music, numbers for arithmetic, ingredients for cooking etc. (XXX 2010, XXX 2022). In this picture then, language is implicated in other generative systems in the sense that these systems also contain the generative device of language. However, even though the idea of a single operation satisfying Principle G in every mental system is methodologically attractive, there is no clear intuition so far that the generative device of language itself is implicated in painting or cooking; indeed, it sounds initially uneasy that the words of a language and the various components of painting combine in the same fashion.

Given the apparently chaotic coverage of the proposed generative mind involving all of speaking Hebrew, singing Tibetan throat music, painting *Guernica*, cooking Biryani and playing chess, it is more likely that, even if some general conception of generativity prevails across domains in terms of Principle G in putting things together, the specific operations vary as domains differ. It is thus plausible to assume that these generative devices are specific to their domains. The assumption yields a generative mind consisting of many generative devices; call it the *multiple-operations* view of Principle G.

The Explanatory Gap

A serious problem with the multiple-operations view is that, for the cited cases, Principle G is most likely to be human-specific in that it does not appear to have been borrowed from pre-human resources. Note that the claim of human-specificity is restricted to the *noted* examples of language, music, cooking, painting etc., each of which admit of unbounded and potentially novel expressions.[5] I am aware that even this narrower claim may be challenged by citing examples of non-human construction of complex objects.[6] However, it is difficult to locate any significant stretch of the generative mind in non-human species, not to mention locating the entire stretch in a single species. Hence, given the limited space of a paper, I will simply assume human-specificity for the suggested coverage of the generative mind. In any case, as we will soon see, the basic explanatory problem is to give a plausible evolutionary account of the vast phenomenon of generativity in whichever species it occurs; in that sense, the basic problem is independent of human-specificity.

The assumption of species-specificity means that the generative mind could not have originated from quantitative modifications of pre-existing functions. Therefore, instead of gradual evolution from non-human species, each generative operation seems to require a *saltation* for its origin. A saltation, which occurs typically in an individual and creates a ‘hopeful monster’, is a sudden and large mutational change from one generation to the next giving rise to emergent biological forms, potentially causing single-step speciation.

In post-Darwinian biology, there is increasing interest in saltational explanations, but they are typically restricted to relatively simple life-forms: polyploid plants, some bacteria, sticklebacks, moths, butterflies, beetles, centipede, roundworm etc., where there is enough genetic knowledge (Minelli et al. 2009, Chouard 2010, Jablonka 2012, Leimar et al. 2012). For higher-order cognitive abilities that concern us here, the required biological understanding has not yet been reached: we do not know what mutations are involved for recursive arithmetic or human language to emerge in a single step. It is not unreasonable to conclude then that (genuine) saltational explanations are currently not available.

It is sometimes claimed that the evolution of more complex organisms is better understood in terms of 'accepted Darwinian mechanisms' involving traits, natural selection, adoption etc. (Roberts et al. 2008). However, since we have assumed species-specificity for the cited generative abilities, standard Darwinian mechanisms of gradual evolution from pre-existing functions appear to be unavailable. Under this assumption then, we face a wide explanatory gap. Indeed, for the concerned abilities, the explanatory gap appears to percolate down. That is, even if there is extraordinary evidence of a pre-human creature from which humans allegedly adopted a higher cognitive function without qualitative modification, understanding the origin of the function in the creature faces exactly the same explanatory gap as with humans. The discomfort is enhanced when many explanatory gaps arise, as in the multiple-operation view, to account for a very large number of cognitive functions of a single species.

Under these circumstances, we may aim for the least explanatory gap in our understanding by thinking of the generative mind as consisting of a single generative device and await further understanding of the device in plausible evolutionary terms.

Earlier, we thought that the understanding of the generative mind may achieve explanatory simplicity if it turned out that Principle G enforces the same generative operation for all mental systems. We can now align that idea with the restriction to a single device for narrowing the explanatory gap. The strategy suggests that it should be possible to identify the (shared) generative part of all mental systems by studying the individual organization of any of them.

In this restrictive context, the study of the generative system of language is almost forced on us because there is a widespread claim that human language is specific to humans (Bickerton 1995, Berwick and Chomsky 2016, Berwick and Chomsky 2019, Tattersall 2019).[7] Adopting this view means that, since the explanatory choice is restricted to just one operation, the generative device of human language already occupies that choice. Hence, I basically examine the organization of language to locate the part, if any, which may be reasonably viewed as the shared generative device satisfying Principle G.

An explanatory advantage with this strategy is that, given the assumption that language is species-specific in its generative capacity, the strategy is likely to ensure the species-specific character of Principle G. Alternatively, given the controversy regarding the evolutionary origin of language, a methodological advantage with the strategy is that the much broader issue of the cognitive divide between human and non-human animals with respect to generativity is brought down to the issue of the origin of the structuring principle of language.

However, the prevalent view among generative linguists is that the structuring principles of language are confined to the domain of language. Within the domain of language, Hauser et al. (2002) suggest a distinction between the narrow faculty of language (FLN) and the broad faculty of language (FLB). FLN contains just the structuring principle of language and FLB contains other aspects of language such as sound and meaning. According to Hauser et al. (2014), FLN 'describes processes that are uniquely human and unique to language.'[8] Even more explicitly, Chomsky (2014, 4) holds that language represents a domain-specific mental faculty, one that rests on structural organizing principles and constraints not shared in large part by other mental faculties.[9] Suppose these claims mean that the structuring operations of language are language-specific.

If so, then the structuring operations of language cannot satisfy Principle G for all domains. Non-linguistic generative systems then can only be governed by non-linguistic structuring operations. As a result, the multiple-operation view of Principle G becomes the only conceptual option. Yet, as we saw, the multiple-operation view is explanatorily unattractive. From this perspective, the only available theoretical course for reaching a well-formed conception of Principle G is to challenge the alleged linguistic-specificity of structuring principles of language.

Merge

At this point, we need to take a brief look at the overall design of language to locate the desired part, if any, that might satisfy Principle G. The design of language incorporates aspects of sound and meaning to generate sound-meaning correlations without bound. The task then is to identify the device that generates unbounded structural relations between sound and meaning. For that purpose, I will adopt the conception of language under the minimalist program in linguistic theory (Chomsky 1995) to see if the proposed conception of Principle G may be extracted out of it. The reason for choosing this theory of language for the project is that it offers a very austere and theoretically salient view of the generative aspect of language with an explicit architecture.[10] The structure of the theory thus provides an opportunity to inspect its components exhaustively to determine if any of them, or some part of them, qualifies as Principle G.

In the minimalist program, the atomic units of language, namely the lexical items (informally known as words), consist of syntactic, semantic and phonetic features: the word *book* contains the sound feature /*bʊk*/, syntactic feature noun, semantic feature artifact. Roughly, the computational system works on syntactic features to combine these units into larger assemblies. Currently, the computational system consists of exactly one combinatorial operation called *Merge*.[11] At some point during computation, the assembled phonetic features are shifted to the sensorimotor (SM) systems to determine how the assembly is to be articulated, and the semantic assembly is transferred to the thought systems to determine the meaning of the semantic assembly; more specifically, the thought systems are characterised as conceptual-intentional (CI) systems. The SM and CI systems are viewed as ‘external’ to the core language system such that the language system *interfaces* with the external systems for each cycle of computation.

Words are thus the input to a computational system, whose sole structuring operation is Merge, and the interfaces are the output of the system. A crucial point is that although the structure of expressions at the interfaces is determined by Merge, the information content of the interfaces flow from words due to what is known as the *Inclusiveness Condition* which states that no new objects are added during computation apart from rearrangement of lexical properties (Chomsky 1995). Since Merge is the only structuring mechanism of language, Merge is the sole candidate, if any, for satisfying Principle G.[12] Holding on to the single operation view of Principle G, the conception of the generative mind requires that the generative mechanism of language be implicated in other mental systems to qualify as Principle G. If Merge is that operation, then Merge must be implicated in a variety of kindred generative systems.

However, even if we think of Merge in isolation as a domain-general device, it is unlikely that Merge works in exactly the same way in each of the otherwise widely disjoint domains. Thus, it is not ruled out that when Merge operates in a specific domain, say language, it works in accordance with the specific properties of that domain. Following this direction, we may think of Merge as a general resource for mental computation whose domain-specific execution possibly varies from domain to domain. While Merge may satisfy Principle G by providing a general computational format for all domains, it may have a 'parametric' form in that how the resources of Merge are used in a domain depends on the conditions specific to the domain.[13] This may be one way of resolving the tension between domain-general and domain-specific conditions while admitting just one operation.

With this perspective in hand, I will now briefly illustrate the suggested abstract character of Merge that leaves some room for variations in accordance with conditions in specific domains. An abstract chracterisation of Merge faces two historical issues. First, with a few exceptions as noted below, Merge has been studied as a generative operation in a specific domain, that is, language. So, definitions of Merge in linguistic theory sometimes carry linguistic properties. Second, theory of Merge is a work in process such that it has undergone several revisions in recent years. Hence, we need to take a quick look at the history of linguistic theory to extract the required abstract conception of Merge.

Earlier, we suggested that Principle G conveys the idea that every generative system contains some operation that takes two existing objects and forms a new one.

In linguistic theory, Merge is an operation that takes two syntactic objects α and β and forms a binary set. Thus,

$$\text{Merge}(\alpha, \beta) = \{\alpha, \beta\},$$

where α and β are either elements from the relevant lexical workspace or objects previously constructed by Merge. In this way Merge exactly satifies Principle G. Moreover, Merge appears to be the simplest combinatorial idea since it is hard to think of a simpler combination than an unordered binary set.

However, in Chomsky (1995) it was said that as Merge constructs the set {α, β}, one of the elements of the set is projected for labeling the set to determine the type of the phrase {α, β}. It meant that Merge itself both generates the set and labels the set. The suggestion leads to a more complicated definition of Merge:

$$\text{Merge}(\alpha, \beta) = \{\alpha, \{\alpha, \beta\}\} \text{ or } \{\beta, \{\alpha, \beta\}\}.$$

This way, Merge identifies the head of a phrase to say, for example, that *the man* is a determiner phrase in which *the* is the head and *man* the compliment. Although authors like Pesetsky (2007) and Berwick (2011) have used projection of heads to characterise musical and metrical structures as well, the requirement seems to be essentially linguistic in character to determine various linguistic objects such as noun phrase, verb phrase, complimentizer phrase etc. Let us suppose so.

Yet, in this arguably linguistically specific formulation of Merge, the simplest idea of Merge in the form of {α, β} is retained and embedded in the more complicated formulation. In any case, I am setting the more complicated definition of Merge aside because Chomsky (2013) used just the simplest form of Merge and viewed projection/labeling separately as a result of minimal search during computation.[14] A detailed discussion of the technical point is beyond the scope of this paper.

Interestingly, similar remarks apply to the even more complicated definition of Merge advanced recently by Chomsky. As noted, proper characterization of Merge in language is currently a hot topic in biolinguistics. For example, authors have suggested various versions of Merge such as Late Merge, Sideward Merge, etc.[15] In this context, Chomsky (2020) suggests a more restricted operation Merge that is defined over (linguistic) workspaces (WS):[16]

$$\text{Merge WS } [\alpha, \beta, \ldots] = \text{WS}' \, [\{\alpha, \beta\}, \ldots],$$

Where '…' contains the rest of the elements in the original workspace WS. Chomsky proposed this restrictive idea of Merge not only to capture the empirical effects of Late Merge, Sideward Merge, etc., but to incorporate some linguistically specific phenomena such as island restrictions. Once again, a detailed discussion of these technical points falls beyond the scope of this paper. But, notice that the simplest form of Merge, {α, β}, has been retained and embedded in the new definition as well. In fact, Chomsky et. al. (2023, p. 106, fn 29) explicitly point out that the fundamental operation in Merge is binary set formation {α, β}.

The presence of {α, β} in any formulation of Merge is not surprising because this form endows Merge with its generative power. In this paper, I am not really concerned with the structural operations of a specific system *per se*, but only with the generative aspect of the system.[17] Therefore, I will hold on to the simplest form of Merge to examine whether Merge is an abstract operation satisfying Principle G in all domains of the generative mind.

Merge is a binary operation that generates an unordered set {α, β} under the familiar extension condition, or no tampering condition, governing sets; so, if γ is merged to {α, β}, it yields {γ, {α, β}}, thus forming a hierarchy. When Merge combines syntactic objects collected from the lexical workspace, it is called *External*

Merge. Merge can also operate with parts of an object already constructed, called *Internal Merge*, {α, {…α…}}; Internal Merge enforces syntactic movement. Thus, regarding the English sentence, *which book the girl has read*, External Merge constructs the structure [[*the girl*] [*has* [*read* [*which book*]]]] after five applications. Given the requirement for English, Internal Merge moves the *wh-* expression *which book* to the clause front to construct the required structure. In human languages thus both External and Internal Merge are in operation, where Internal Merge is driven by lexical features such as *wh-*.

Spread of Merge

We saw that the Inclusiveness Condition governs the flow of linguistic information from the lexicon to the interfaces. The computational operation itself does not contribute to the information contained in the resultant structure. It appears as though the computational part is *just there* to endow the total system with generative power, irrespective of the information flowing from the lexicon. The design encourages the strong intuition that Merge may not be intrinsically connected to the domain-specific features of language (XXX 2010, XXX 2022).[18] When α, β are linguistic objects, no doubt Merge generates linguistic structures: {*saw* {*the man*}}. Yet, Merge itself is not domain-specific because the function of Merge is to generate sets once the lexical storage from any relevant domain supplies symbols in a workspace; Merge generates sets blindly (XXX 2010, XXX 2022).

The versatility of Merge accrues from the idea of a set. Just as there are sets of tables and tall mountains, there are sets of imaginary quantities {√-1, √-2}, impossible objects {round square cupola, circular triangle}, even unrelated objects {√-1, circular triangle} and visual objects {☹, §}. These are all legitimate sets because their members are represented in legible symbols; these sets are products of Merge. To form a set, objects have to be identified as falling under (abstract) categories that are identified in terms of a symbolic form. The formation of a set by Merge thus requires symbols and in that sense Merge is a very abstract operation of the mind (XXX 2022). Just to get some flavour of versatility of Merge, I will now very briefly illustrate the domain-generality of Merge, along with its domain-specific application as noted, with some examples (XXX 2022 for details).

It has been suggested that Merge operates on symbols like x and /, which may stand for strong or weak metrical stress, to generate metrical structures such as x / x / x / x, as in *tell me not in mournful numbers* (Berwick 2011). For our purposes, the most interesting feature of metrical stress-structures is that they do not contain the lexical features of human language. As Berwick (2011) states, it is an 'impoverished' syntax 'with no lexical features and no associated semantics; without features, there is no possibility of internal Merge and the movement that we see in ordinary syntax'. Berwick's suggestion is somewhat misleading because it implies that Internal Merge can only operate in a system which contains the lexical features of language; in effect, it means that Internal Merge is available only in language. This seems to be a stipulation. A more plausible view is that the logical option of Internal Merge is simply withheld in the domain of metrical stress because the 'lexicon' of metrical stress does not enforce this option. That may not be the case in other domains in which Merge operates without linguistic features.

The system of arithmetic is a case in point. It is well-known that the numerical system from 0 onwards may be viewed as a product of Merge. According to Chomsky (2007), Merge operates on a single lexicon 0, and the result is the successor function: 0, {0}, {{0}}, {{{0}}}… In digits, we may call {0} as 1, {{0}} as 2, and so on, to enumerate the familiar series of ordinals. In that sense, the digital sequence 1, 2, 3, …

represents the structure generated by Merge. In more recent work, Chomsky (2020) suggests that the ordinal series is generated by Internal Merge as it operates repeatedly on its own earlier product {0}. If so, then Internal Merge not only operates in arithmetic without linguistic features, Internal Merge is the *only* generative operation in arithmetic.

Another difference beween the application of Merge in arithmetic and language is that, while the generative structure in arithmetic strictly obeys the x-over-x principle such that Merge keeps operating on the same syntactic category, structures in language typically require incorporation of different syntactic categories: [$_{CP}$ *Allegedly*, [$_{TP}$ *John will* [$_{vP}$ *deny* [$_{DP}$ *the very possibility* [$_{CP}$ *that* [$_{TP}$ . . .]]]]]] (Arsenijevic' and Hinzen 2012).[19] It follows that, once we delink the idea of Merge from its domain-specific operations in language, the basic operation could be viewed as more abstract than any of its specific local applications, which are likely to be partly sensitive to the domain concerned.

As to the domain-general character of Merge as it applies to arithmetic, there is interesting evidence that arithmetical functions remain in place despite severe impairment in some components of language (Varley et al. 2005). The study was conducted on three patients who suffered from difficulty in processing phonological and orthographic properties of lexical items including number words. These patients also failed to perform on language-specific grammatical tasks such as tracking argument-structure and identifying clauses. The authors report that, otherwise, all three patients solved mathematical problems involving recursion and structure-dependence when arithmetical problems were posed in digits. For example, the patients showed high level of competence in resolving bracketing issues. This result indicates that the computational procedures of arithmetic, when conducted in digits, remain unaffected despite impairment affecting number-words. Hence, insofar as Merge operates in arithmetic to generate digits and arithmetical functions, the operation of Merge remains unaffected by language-specific impairment.[20]

Similar variations in the application of Merge may be seen in the domain of music. The generative character of human music was recognized for ages. Every musical system consists of a small set of notes and syllables as units that enter into computation. These units are compiled over and over again to generate progressively complex objects such as chords, scales, modes, phrases, passages, movements, and so on. The generation of non-lexical complex objects is unbounded, hierarchical, and countable (Brown 2000; Merker 2002; Fitch 2006). Thus, some researchers have in fact proposed that language and music share structural parallels such as hierarchical organization, recursive operations, and long-distance dependencies (Patel 2003; Rohrmeier 2011). In a review of neural research on this topic, Faroqi-Shah et al. (2019) report that both musical and language syntax are processed by the same domain-general cognitive mechanism located in the left frontal lobe, but rely on distinct neural representations in the temporal lobes.

These experimental observations seem to match the more formal studies on language and music in which the operation Merge is increasingly proposed in the literature to explain the unbounded hierarchical character of musical forms (Katz and Pesetsky 2011; XXX 2010; Mukherji 2013; Hauser and Watumull 2017; XXX 2022).[21] Katz and Pesetsky (2011) in fact propose an ‘identity-thesis’ for language and music. In their framework, Merge generates complex musical structures with repeated application of External Merge. The authors show how the opening part of Mozart's piano sonata K. 331 may be generated from applications of Merge, the details of which I set aside (XXX 2010). It appears that the idea of using External Merge to explain musical progression extends to other genres and forms of music (Mukherji 2013).

Katz and Pesetsky (2011) also suggest that Internal Merge operates in music to trigger movement of chords to form cadences. However, as with melodic stress and arithmetic, the lexicon of music does not contain the features that drive movement in language. Hence, even if cadential movement in music is driven by Internal Merge, the movement is probably guided by music-internal constraints such as adjacency condition for phrasal closure (Mukherji 2013). If so, then operations in music consist of both External and Internal Merge like language, but unlike language, Internal Merge in music obeys music-internal structural conditions. In any case, in view of the marked differences between language and music in their domain-specific properties (Patel 2003; Peretz et al. 2015; Fedorenko and Varley 2016; Faroqi-Shah et. al 2019), Katz and Pesetsky's 'identity thesis' appears to be unwarranted.

The preceding comparative study of language, metrical stress, arithmetic and music suggests at least two ways in which the abstract character of Merge is not language-specific. First, within the domain of language, abstract Merge is basically independent of the language-specific parts of the domain, namely the lexicon and the resulting interface constructions; hence, there is nothing linguistically specific in Merge itself. Second, the basic format of Merge appears to be available to a variety of non-linguistic domains although the resources of Merge are utilised in somewhat different ways by these domains. We may thus think of abstract Merge as a domain-general operation of the generative mind.

How general? Although there have been occasional studies using Merge or Merge-like operations in apparently non-linguistic domains such as concept formation (Thornton 2016), kinship relation (Hale 1966, Jones 2010), morality (Mikhail 2007, Hauser and Watumull 2017), and the like (XXX 2022), no formal theory is yet available in any of these domains to my knowledge; to that extent, we do not know how far Merge-based explanations extend to other mental domains beyond those already discussed.[22] Hence, filling the gap between the conceptual demands of Principle G and the current empirical reach of Merge is work in progress.

Extending Principle G

We have so far assumed that Principle G covers generative systems, such as language, music, and arithmetic, as we find them in the current state of humans. After the preceding survey, the question arises whether Merge may be viewed as applying to even more distant non-linguistic domains, perhaps even to recent hominid ancestors of humans. If so, then the coverage of Principle G extends to these domains as well. A few brief remarks on this point may be in order by way of conclusion.

Some archaeological findings illustrate the point. Archaeologists sometimes explain the making of stone-tools by earlier hominids in terms of hierarchical structures (Nowell and Davidson 2010; Stout 2011; Joordens et al. 2015). For the Late Acheulean shaping of stone-tools, Dietrich Stout draws tree-diagrams to show at least six levels of hierarchic operations such as hammer stone selection, platform preparation, percussion, target selection, complex flake detachment, etc. On the basis of such results, Uomini and Meyer (2013) hold that the 'neural processes underpinning language' are the same as those 'underlying Acheulean stone-working techniques' dating back to 300K years. The authors are clearly suggesting that the same, or very similar, structural principles underlie the mental systems of language and Acheulean stone-working.[23]

There are two problems with such claims. First, the structural operations of language could not have been available so early in hominid evolution because language emerged about 150K years ago (Tattersall 2019). Second, if some other language-like operation applied in the Late Acheulean case with the generative features of unboundedness and novelty, it faces the explanatory gap with respect to the pre-linguistic ancestor. However, since in our analysis, the basic operation Merge is no longer viewed as linguistically specific, it is a plausible conjecture that Merge, as the simplest computational device, applied in the Acheulean case. Also, since the origin of Merge already carries the problem of explanatory gap, the problem will not multiply if Merge now applies to a remote pre-linguistic case.

Moreover, the postulation offers the explanatory advantage that it places the emergence of Merge before the emergence of language in hominid evolution. Therefore, Merge may now be used to explain some hitherto unexplained aspects of language evolution itself. As Ian Tattersall (2016, 2017, 2019) has reported over the years, the human species developed two novel capacities: the ability to 'process information symbolically to form ideas which are then used to form structured articulate language'. In more familiar terms, humans developed the abilities for word-formation and word-combination. The language system formed when Merge started operating in the environment of words to generate structured expressions at the interfaces.

A word has a sound and a meaning; so we can think of a word as a unit of sound-meaning correlation (XXX 2021).[24] We saw that the minimalist program of linguistic theory holds that elements of sound and meaning are stored in SM and CI systems, perhaps prior to the emergence of structured articulate language (Chomsky 2012, p.14-5). Suppose the SM systems contain the elements of sounds, gestures etc. and the CI systems contain conceptual elements, perhaps in the form of perceptual images.

The point is, even if the SM and CI systems are individually in place, words are still not in place because word-formation requires that elements of SM and CI systems have combined. For example, in order for the word *banana* to form, the sound /banana/ and some conceptual element BANANA must combine.[25] According to the Chomskyan picture of a narrow faculty of language (FLN), the generative operation of language combines words *the* and *man* to form the structured expression *the man*. The question is, which combinatorial operation combines elements from SM and CI systems to form words? Words are rather weird objects in that they result out of elements from two entirely disjoint domains, such as sounds and concepts. Further, word-formation can be typically unbounded due to inflections: *verb*, *verbal*, *verbalize*, *verbalization* etc. Moreover, as linguists point out, words such as *unlockable* are hierarchical and structurally ambiguous (Pollatsek et al. 2010, Hauser et al. 2014).

It thus appears that word-formation requires an unbounded and hierarchical operation. In sum, word-formation falls under Principle G. If there is a separate, domain-specific operation for word-formation that meets these conditions, we face a new problem of explanatory gap. However, the simplest Merge already satisfies the required properties of domain-generality and unboundedness. It is therefore plausible that Merge forms the structure {/banana/, BANANA}, which is spelt out as *banana*.[26] In this way, Merge could be satisfying Principle G in both word-formation and word-combination to bring about the human language system.[27]

Endnotes

[1] It may be instructive to find out how the conception of the generative mind relates to common conceptions of the human mind; however, in this paper, I will not be concerned with that issue (see XXX 2022).

[2] See XXX 2022 for some discussion on relevant computation.

[3] Classical philosophers such as Rene Descartes did not ascribe mentality to animals at all because they lacked the linguistic ability (Clarke 2003, Chomsky 2015, XXX 2022).

[4] However, recent discovery of symbolic objects at much earlier dates involving other hominids seems to refute the alleged link with human language (Stringer 2011, Balari et al. 2011, Barceló-Coblijn and Gomila 2012 etc.).

[5] There is some dispute whether non-human animals display any combinatorial ability at all (Penn et al. 2008, Truswell 2017, Townsend et al. 2018, XXX 2022).

[6] To that end, one could cite insect locomotion, primate calls, bird songs, beaver nests, and the like. But then it needs to be shown first that the suggested non-human examples do meet the conditions of unboundedness and novelty we routinely find in humans. See XXX 2022 for extensive discussion and references on this issue.

[7] The claim is controversial (Progovac 2015, Planer and Sterelny 2022). I will hold on to the human-specific view of language because, for expository simplicity, I will approach the generative properties of language via a theory of language that upholds this view (Berwick and Chomsky 2019). I do not discount the possibility that the topic of generativity may be pursued with other conceptions of language.

[8] In Hauser et al. (2002) it was suggested somewhat differently that 'FLN only includes recursion and is the only uniquely human component of the faculty of language [but] may have evolved for reasons other than language.' Thus, it was

granted that, even though recursion is uniquely human, it might have been functioning in other non-linguistic but human-specific domains. See Hauser and Watumull (2017) for more; see XXX 2022 for criticism of Hauser and Watumull (2017).

[9] For records, there are at least two occasions in which Chomsky thinks of the principles of language applying elsewhere. Chomsky (1988, 169) suggests that just the 'mechanism of discrete infinity' applies to arithmetic after 'eliminating the other special features of language.' In Chomsky (2000, 27), he speculates that even a nonhuman organism might be endowed with the structuring principles of human language for using it in locomotion (XXX 2022).

[10] As we will see, the minimalist approach contains just one combinatorial operation, Merge. In contrast, an approach such as Progovac (2015) contains at least the following: Adjoin, Conjoin, Concatenate, Coordinate, link, proto-Merge, Merge, Parataxis, Hypotaxix etc.

[11] Sometimes, another operation called *Parallel Merge* (PM) is introduced to explain the construction of sequences and adjuncts (Chomsky 2020). However, PM is barred in Chomsky et al. (2023, p. 111, fn. 55).

[12] According to Berwick and Chomsky (2019), Merge is human-specific as there are no half-Merges or demi-Merges elsewhere in nature. As noted, human-specificity is not the real issue; the explanatory gap is.

[13] Chomsky (2007) seems to be making a similar proposal in suggesting that even if Merge is appropriated from other systems, 'there still must be a genetic instruction to use Merge to form structured linguistic expressions satisfying the interface conditions.' See Satik (2022) on how Merge may have adapted to more specific syntactic conditions of language.

[14] Van Gelderen (2024, p.2) suggests that the simplest form was retained as the only definition of Merge at least upto Chomsky et al. (2019).

[15] See Chomsky (2019) for criticism of these extensions of Merge.

[16] MERGE in capitals was used in Chomsky (2020) to distinguish it from earlier formulations of Merge. However, in Chomsky et al. (2023), the original *Merge* was used for the new definition.

[17] Following the preceding discussion on domain-specific implementation of otherwise domain-general principles, it could be that the new WS-based definition of Merge reflects properties of linguistic lexicon. A detailed discussion falls out of the scope of the paper.

[18] Thus, in contrast to language-specific rules, construction-specific rules, and general linguistic principles, Merge may be viewed as a purely computational principle of language, (XXX 2003, XXX 2010). I am setting aside the issue of whether purely computational principles belong to what Chomsky (2005) calls the *third factor*, principles that reflect general laws of nature (see XXX 2010).

[19] The authors suggest that the generative operation in arithmetic is not Merge at all, if the operation in language is. Their suggestion holds only if Merge is viewed as language-specific, which we are challenging.

[20] Some authors such as Friederici et al. (2011) suggest that heirarchical structures in language and arithmetic are processed in somewhat different locations in the brain showing perhaps that they involve distinct operations. However, more recent studies suggest that linguistic and arithmetic processing occur in distinct brain areas with a domain-general operation available to both (Fedorenko and Varley 2016).

[21] Since Merge operates on symbols, it is sometimes questioned whether musical elements are symbolic at all; see XXX 2010 and XXX 2022 for extensive discussion.

[22] However, applying Merge-like operations to explain non-hominid behaviour such as ordering of objects by Capuchin Monkeys (McGonigle et al. 2003), classification of kinship by Japanese Macaques (Schino et al. 2006), breaking of nuts by chimpanzees (Fujita 2014) etc. is questionable because the crucial factor of set of symbols is unlikely to be available in the cognitive architecture of these animals (XXX 2021, XXX 2022).

[23] See Planer and Sterenly (2021) for an attempt to explain evolution of language from ancient tool-making, see Progovac (2021) for critical discussion.

[24] As noted, a lexical item also contains syntactic features which categorises words as nouns, verbs etc. The more complex issue of how syntactic categories are introduced in the lexicon falls outside the scope of the paper.

[25] We assume of course that the elements /banana/ and BANANA are already stored somehow in the SM and CI systems. According to the Galilean challenge, at least the construction of the complex sound /banana/ needs to be explained. We set the problem aside for now (see XXX 2022).

[26] Since the sound-part of the structure is language-specific (English in this case), words are language-specific. In that sense, as Thomas Wasow (1985) pointed out, knowing a language basically means knowing the words of a language; the generative part of language comes for free by human mental design.

[27] If we assume that the conceptual element in BANANA is just a perceptual image, the merged structure {/banana/, BANANA} possibly signals how percepts grow into concepts in terms of words. A detailed discussion is beyond the scope of the paper (XXX 2021, XXX 2022).

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